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AI/ML Capstone Project

D1, Project Proposal (5% of final mark)

Due: Week 4 (softcopy on SLATE, hardcopy to your professor in class)

Name 1: Afolabi Adesina	Student Number 1: 991808102
Name 2: Evans Frimpong	Student Number 2: 991771602
Name 3: Xian Qin	Student Number 3: 991578381
Name 4: Yanan Yang	Student Number 4: 991841037

Instructions:

- Please work on the following deliverable with your group.
- Prepare your deliverable report as discussed in class.
- Type your report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload your report in PDF format on SLATE. Team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stappled) of your deliverable to your professor in class in Week 4

Q1: Which sector driving modern economies are you considering for your capstone project (i.e., agriculture, education, energy, entertainment, finance, government, health care, insurance, manufacturing, NGO). Select the sector that is mostly relevant to your capstone. Briefly explain your sector selection and what experience you and your group have in this area. Answer this question in one paragraph.

For this project, we chose the highway and transportation industry. Our project focuses on the safety of the roads that leads the everyday human to and from their destination and the application of artificial intelligence to provide a solution that maximizes the choices of safety. As pertaining to the reason of choice of this project, our team consists of individuals with statistical and data analytics knowledge together with the technical aspect of application production. The combination of these skills gives us the ability to maneuver around a project such as the one we have chosen. There are also a wide range of documentation on the analysis of road safety and availability of the datasets that will be needed in this project.

Q2: Identify a real-world problem that you and your group are interested to explore. Explain the identified “problem” in as much detail that you can. Provide citable data or context to prove that this problem exists and that it matters. Do not forget the famous saying that: “a problem well stated, is a problem half-solved”. Answer this question in two to three paragraphs.

The problem identified is that most GPS applications that recommend routes to and from a destination only do so on the basis of recommending the fastest route without necessarily considering the riskiness of the roads[1]. Our project with the help of artificial intelligence, is going to be built with the component of safety attached. We believe that factors such as weather, historical accidents on a given road among other conditions should be taken into account when recommending routes. There have been some great introduction to safety options for route recommendations in the recent years [2] and this project is going to give a look into our perspective when it comes to approaching this issue and how our solution could be a build up to what has already been established.

Q3: Identify the prospective clients and stakeholders that the real-world problem that you explained above is targeted for. What are their specific requirements? How are they currently solving this challenge without your AI/ML solution? Support your ideas with reliable market research data. Answer this question in one paragraph.

The target clients for the Ontario “Smart-Shield” include daily commuters on the 400-series highways, commercial trucking fleets, and the Ministry of Transportation Ontario (MTO). Their specific requirement is simple but urgent. They need real-time, hyper-local hazard warnings during bad weather. They need to know if a specific road segment is safe to drive on

right now. Currently, drivers rely on standard navigation apps like Google Maps, Apple Maps, or Waze. However, these solutions only optimize for travel time and distance. They suffer from a major “Static Data Gap.” They cannot detect sudden, dynamic environmental hazards like black ice, wet slush, or sudden whiteouts [1]. Because of this gap, drivers do not know a road is dangerous until they physically hit an accident zone or a sudden traffic jam. Market research shows a massive annual spike in multi-vehicle collisions during winter months. This proves that current GPS tools fail to actively protect drivers because their underlying engines treat safety purely as an afterthought to speed [2]. Our model solves this challenge by ranking routes based on actual environmental risk, not just ETA.

Q4: Explicitly justify why an AI/ML solution is needed here rather than applying standard heuristics or traditional software engineering alternatives? What are the benefits of using AI/ML to tackle this real-world problem? Answer this question in one paragraph.

AI/ML solutions are needed to collect, process and analyze real time data from camera feeds, alert message or text data from government about road work/constructions with weather data on the go. Traditional software engineering approach relies on rule-based logics which cannot process large data and unstructured data successfully. AI/ML solution provides the scalability option, where several types of data can be analyzed simultaneously using natural language processing for text, neural network for find optimal solution based on number of factors and then logistics regression for the final decision.

Q5: Explain how AI/ML can be integrated into the real-world problem you identified above. Support your ideas from existing literature and/or industry benchmarks. Answer this question in two to three paragraphs.

To make this work AI/ML solution applied using, the system uses natural language processing to read emergency text alerts [3] and a convolutional neural network to visually spot surface hazards [6]. These insights are then combined with real-time weather data using a logistic regression model to generate a segment-specific safety score, relying on proven multi-modal fusion strategies [7]. By focusing on catching high-risk conditions rather than just general accuracy [5], the tool proactively warns drivers about severe dangers before the first collision occurs.

Q6: The AI/ML capstone project expects groups to work with datasets. In general, groups can train, validate, and test an AI/ML model of their choice. Alternatively, groups may consider customizing an SLM or LLM with specific or proprietary data using techniques like finetuning or RAG. Provide some insights on where will your data come from? Is the data available in open-source, do you need to scrape, generate it on your own, or synthetically simulate it? Provide explanation on licensing, data privacy, and the volume

sufficiency needed for training, model adaptation or context injection. Also, is the considered dataset properly labelled or not and what implications will this have on your overall work. Answer this question in three to four paragraphs

Our primary data source will be government public dataset. There are three kinds of data we mainly need: traffic collisions, weather, and road conditions. For the traffic collisions, we found the dataset such as Police Annual Statistical Report - Traffic Collisions [9] and the Traffic Collision Dashboard (2019 to 2022) [10]. These open-source datasets are legal, reliable, and authoritative. Also, we do not need to worry about the licensing and privacy issues, since they are from the government, but we will still double check with our own regulations and policies to make sure they are safe to use.

For secondary, we are planning to get real-time data from surveillance cameras to make our prediction more accurate. There are two kinds of data from these cameras, videos, and photos. We will capture the meaningful frames from the videos, and the photos from certain government websites, to analyze the pictures to recognize accidents, weather, and road conditions. The volume of the dataset we have found is sufficient. Take the Police Annual Statistical Report - Traffic Collisions [9] for example, has approximately 768,000 rows of collision records; it is statistically significant to train, test and optimize our AI models.

What features to use are the key questions to make a good model. To improve our analysis, we will explore more features such as environment features, weather patterns, road construction schedules, local special events, and buildings along the roads. Some of these factors will be collected from structured open-source APIs; others will need to be processed. Unstructured data will need a lot of computing power and time. We need to build a computer vision pipeline to process real-time surveillance videos and photos. Our pipeline will recognize which frames are necessary, useful, and legal to use. Also, we need to optimize it so that it will work efficiently and accurately enough to provide sufficient real-time data we need.

Our main historical dataset has lots of key labels we need, such as time, location, and severity. We can get the weather condition of that location from weather reports in that area. Then we will clean, process the data, and train our model based on it. After we get the real-time data from our pipeline, no matter if it is from the videos or text or direct structured datasets, we can make predictions based on it using our model. We will also update our dataset and model periodically to keep our system more reliable and accurate. The risks we may face. First, is no usable real-time data. If we can get access to government APIs, datasets, and websites. As for our research and tests, it is challenging but still feasible. We will find more sources and technical skills to access them. Second is the computing power limit. We are considering using cloud computing services if necessary, and we will keep optimizing our algorithms to reduce the power needed to make our system work efficiently.

Q7: An AI/ML technology stack is the generally composed of the infrastructure layer (i.e., HW/SW), data layer (i.e., injection, cleaning, storage), framework and model layer (i.e., TensorFlow, PyTorch, LLM/SLM, fine-tuning/RAG), ML Ops layer (e.g., Hugging Face, Docker), and the application layer (e.g., APIs, UI, Streamlit). Identify a preliminary technology stack suited to tackle

the AI/ML challenge of your capstone project. Answer this question by providing a brief sentence for each AI/ML technology stack.

- **Infrastructure Layer:** Use Python 3.10 or higher versions to train the models, the computational process will be on a local CUDA-enabled GPUs or cloud-based environment, for example, Google-colab, especially train a CV model and handling heavy text datasets.
- **Data Layer:** The accident datasets collected recent 5 years, the real-time coordinates data and times fetched from the Google Directions API [1], and the weather and road conditions data collected from the Ontario 511 system.
- **Framework Model Layer:** NLTK[3], Scikit-Learn, TF-IDF, OpenCV[4], DNN, Random Forest, and Logistic Regression to predict the accident rates and calculate the final segment risk scores.
- **MLOps Layer:** ML models is designed to run via asynchronous background (e.g., Docker containers) to pre-compute risk scores, in case any delay during user searches.
- **Application Layer:** The final system will be deployed as a standalone web application using JavaScript and will consider making it a web plugin or API extension for Google Maps[1]. So once users input the origin location and destination on the website, it will show the ranked routes color-coded by safety scores.

References

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